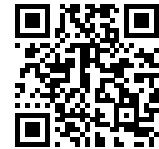


Vishal Khan

AI / Machine Learning Engineer

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[LinkedIn](#) | [GitHub](#) | [AI Twin — live demo](#)



[Chat with my AI Twin](#)

Right to work in the UK. Part-time now; full-time from February 2027; then 18 months unsponsored. Eligible to switch to Skilled Worker as a new entrant thereafter — no sponsor licence required to hire me.

Four years in finance and data engineering before ML. I ship LLM systems to production: a live RAG assistant on **FastAPI**, **ChromaDB** and **Azure OpenAI** with intent-boosted retrieval, SSE streaming, tool calling, prompt-injection defences and **390 tests** in CI; and a multi-agent pricing system over SentenceTransformer embeddings. Previously built **Databricks** pipelines over millions of telecom records at Teleperformance, cutting query time and infrastructure cost by **~20%** each. MSc AI & Computer Science at Northeastern University London, completing January 2027; prior MSc thesis benchmarked DQN, A2C and PPO for dynamic pricing.

SELECTED ENGINEERING WORK

AI Digital Twin — Production RAG Assistant

[Live](#) | [Source](#)

- **OBJECTIVE:** Answer recruiter and interviewer questions from grounded, cited evidence, including live GitHub data a static knowledge base cannot hold.
- **APPROACH:** FastAPI, ChromaDB and Azure OpenAI with **8 LLM-callable tools** and three answer modes, each with its own citation policy. Source-aware retrieval applies per-source distance thresholds before preference boosting, so a preferred source wins near-ties but cannot pull an irrelevant chunk into the prompt; follow-ups fold the last two turns into the embedding query, holding subject continuity without a second LLM call. Auto-ingested GitHub READMEs are delimited as untrusted data against prompt injection. **React 19** and **TypeScript** front end streaming over SSE.
- **RESULT:** Live on **Azure Container Apps** after migrating off Railway, with a GitHub Actions to GHCR image pipeline at **~\$4.70/month**. **390 tests** (276 backend, 114 frontend) under mypy strict, ruff and oxlint in CI; injection defences regression-tested across all three modes.

Multi-Agent Pricing System

[Source](#)

- **OBJECTIVE:** Estimate product prices from unstructured descriptions, and test whether a fine-tuned open model can match frontier LLMs on the task.
- **APPROACH:** Ensemble of specialist agents with a coordinator reconciling their outputs; RAG over SentenceTransformer embeddings in Chroma; frontier agent migrated from the OpenAI SDK to Azure OpenAI, resolving a stale-module authentication failure. Benchmarked **16 models** against a human baseline on one harness — classical ML through XGBoost, five frontier LLMs, and fine-tuned open models.
- **RESULT:** Best model, a fine-tuned 4-bit **Llama 3.2**, cut error **62%** below the constant baseline and beat every frontier model tested, including **GPT-5.1** by **11%** — moving the same base model from worse-than-baseline to best-in-class through fine-tuning alone.

EXPERIENCE

Teleperformance Global Business Pvt. Ltd. [\[letter\]](#)

June 2024 – Jan 2026

Data Engineer

- Built and maintained **Databricks** pipelines processing millions of telecom records (CDRs, billing, agent performance), serving a data engineering team of **50+** across multiple regions.
- Migrated on-prem SQL Server schemas to Databricks, improving query performance by **~20%** and cutting infrastructure cost by **~20%**.
- Owned **9 pipelines** end to end on Databricks Workflows, translating ambiguous stakeholder requirements into scheduled jobs that retired **30+** recurring manual reports.

Avant Garde Clean Room & Engg. Solutions Pvt. Ltd. [\[letter\]](#)

Aug 2021 – Dec 2023

Executive — Commercial & Sales Analytics

- Built ML sales forecasting models achieving **10–20% MAPE** on monthly revenue projections, used daily for commercial planning.
- Turned ad-hoc commercial questions into recurring Python and SQL reporting workflows the team relied on.
- Rebuilt revenue and collections reporting in Excel (macros, VBA), cutting the monthly cycle from **~2 hours** to **~30 minutes**.

Earlier: finance, accounts and taxation — [Aryan Care Pvt. Ltd.](#) (2021) and [PJRJ Associates](#) (2018–2020). Statutory filings, audit support, reconciliation.

TECHNICAL SKILLS

Languages	Python, SQL, TypeScript, JavaScript, R, HTML/CSS
LLM & Agents	RAG, prompt engineering, agentic systems, MCP, LangChain, tool calling, Azure OpenAI, prompt-injection defence, evaluation harnesses
Machine Learning	Deep learning, reinforcement learning, classification, regression, time-series forecasting, NLP, feature engineering, model evaluation
Libraries	PyTorch, TensorFlow, Keras, scikit-learn, SentenceTransformers, Pandas, NumPy, Statsmodels, Matplotlib, Plotly
Engineering	FastAPI, Pydantic, ChromaDB, REST, SSE streaming, React 19, Vite, Tailwind, Docker, Azure Container Apps, GitHub Actions CI, GHCR, Vercel, Railway, Streamlit
Testing & QA	pytest, vitest, Testing Library, mypy (strict), ruff, oxlint
Data & Cloud	Databricks, Databricks Workflows, Azure, AWS, ETL pipelines, pipeline orchestration, SQL Server, MySQL, MongoDB, Neo4j
BI	Power BI, Tableau, Excel

EDUCATION

Northeastern University London, UK [cert]	<i>Jan 2026 – Jan 2027</i>
MSc Artificial Intelligence & Computer Science 95/100 in Programming for Data Applications	
Liverpool John Moores University, UK [cert]	<i>Mar 2023 – Mar 2024</i>
MSc Data Science	
IIIT Bangalore, India [cert]	<i>Jan 2022 – Jan 2023</i>
Executive PG Programme in Data Science	
University of Delhi, India [cert]	<i>Nov 2018 – Nov 2021</i>
BCom (Commerce)	

RESEARCH

MSc Thesis: Reinforcement Learning for Dynamic Pricing in E-commerce [pdf]	<i>LJMU, 2024</i>
▪ OBJECTIVE: Test whether RL agents outperform rule-based pricing policies in a simulated e-commerce environment. ▪ APPROACH: Built a custom Gym-style environment on real e-commerce data, then trained and compared DQN, A2C and PPO agents across multi-episode simulations. ▪ RESULT: DQN produced the most stable profit curve; wrote up the comparative analysis as the dissertation.	
Current dissertation (Northeastern London, due 2027): agentic AI, focused on agent reliability and evaluation tooling.	